

RMO– 2015(MUMBAI)

Max Mark – 100

Time – 4 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions.
- Each of questions 1,2,3 carries 10 points. Each of questions 4, 5, 6, 7, 8.
- Answer to each questions starts on a new page. Clearly indicate the question number.

1. Let $ABCD$ be a convex quadrilateral with $AB = a$, $BC = b$, $CD = c$, $DA = d$. Suppose $a^2 + b^2 + c^2 + d^2 = ab + bc + cd + da$, and the area of $ABCD$ is 60 square units. If the length of one of the diagonals is 30 units, determine the length of the other diagonal.
2. Determine the number of 3-digit numbers in base 10 having atleast one 5 and atmost one 3.
3. Let $P(x)$ be a non-constant polynomial whose coefficients are positive integers. If $P(n)$ divides $P(P(n) - 2015)$ for every natural number n , prove that $P(-2015) = 0$.
4. Find all three digit natural numbers of the form $(abc)_{10}$ such that $(bca)_{10}$, $(cab)_{10}$ and $(abc)_{10}$ are in geometric progression. (Here $(abc)_{10}$ is representation in base 10).
5. Let ABC be a right angled triangle with $\angle B = 90^\circ$ and let BD be the altitude from B on AC . Draw $DE \perp AB$ and $DF \perp BC$. Let P , Q , R and S be respectively the incentres of triangle DFC , DBF , DEB and DAE . Suppose S , R , Q are collinear. Prove that P , Q , R , D lie on a circle.
6. Let $S = \{1, 2, 3, \dots, n\}$ and T be the subset all triples of subset of S , say (A_1, A_2, A_3) such that $A_1 \cup A_2 \cup A_3 = S$. Determine, in terms of n , $\sum_{(A_1, A_2, A_3) \in T} |A_1 \cap A_2 \cap A_3|$, where $|X|$ denotes the number of elements in X . (For example, if $S = \{1, 2, 3\}$ and $A_1 = \{1, 2\}$, $A_2 = \{2, 3\}$, $A_3 = \{3\}$ then one of the element is $T = (\{1, 2\}, \{2, 3\}, \{3\})$).
7. Let x, y, z be real numbers such that $x^2 + y^2 + z^2 - 2xyz = 1$. Prove that $(1+x)(1+y)(1+z) \leq 4 + 4xyz$.
8. The length of each side of a convex quadrilateral $ABCD$ is a positive integer. If sum of the lengths of any three sides is divisible by the length of the remaining side then prove that some two sides of the quadrilateral have the same length.

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